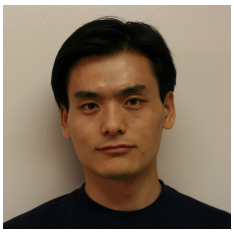

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Education

1999 - 2004 **Ph.D.** in *Experimental Condensed Matter Physics*, Michigan State University, USA
Supervisor: Prof. Simon J. L. Billinge

1994 - 1999 **B.S.** & **B.E.** in *Physics* & *Computer Science*, Univ. of Science & Technology of China
Supervisor: Prof. Liren Lou

Profession

2010 - now **Assistant Professor**, Department of Physics, George Washington University

2007 - 2010 **Research Fellow**, Lab of Physical & Structural Biology, National Institutes of Health
Mentor: Dr. V. Adrian Parsegian

2004 - 2007 **Postdoctoral Associate**, School of Applied & Engineering Physics, Cornell University
Mentor: Dr. Lois Pollack

Presentations (invited, since joining GW)

1. 10/2011, MS&T 2011 Conference, Columbus, OH. *Tightly packaged DNA in a test tube and in a nanobio-container*
2. 09/2011, Biophysics Seminar, University of Maryland, College Park, MD, *Molecular force of nucleic acids assemblies*
3. 02/2011, International Conference Nanobiophysics/Nanochemistry, Sharm el Sheikh, Egypt. *Tightly packaged DNA in a test tube and in a nanobio-container* (postponed)
4. 10/2010, Material Science and Engineering Program, Columbia University, New York, NY. *Electrostatic, hydration, and bending forces in compact DNA assemblies*
5. 10/2010, Department of Physics and Astronomy, Howard University, Washington, DC. *Electrostatic, hydration, and bending forces in compact DNA assemblies*
6. 08/2010, Denver X-ray Conference, International Centre for Diffraction Data, Denver, Co. *Force measurement of DNA with Pair Distribution Function*

Presentations (contributed, since joining GW)

1. 02/2012, Biophysical Society 56th Annual Meeting, San Diego, CA. *Quantify cation-dependent forces between nucleosomes*
2. 02/2012, Biophysical Society 56th Annual Meeting, San Diego, CA. *Sequence and Structure Dependent DNA-DNA Interactions*
3. 02/2012, Biophysical Society 56th Annual Meeting, San Diego, CA. *Elucidating mechanisms of nucleosome aggregation: low resolution structure studies of a variety of nucleosome core particle constructs as a function of changing ion*

4. 02/2012, Biophysical Society 56th Annual Meeting, San Diego, CA. *Elucidating the role of ions in DNA condensation: measurements of the ion atmosphere surrounding condensed DNA pellets using inductively-couple plasma atomic emission spectroscopy*

5. 07/2011, Biophysical Society Conference "Dynamic DNA Packaging Across Kingdoms: Chromatin & Beyond", Asilomar, Pacific Grove, CA. *Probing DNA structure and energetics in vitro and in vivo with x-ray scattering*

6. 07/2011, Biophysical Society Conference "Dynamic DNA Packaging Across Kingdoms: Chromatin & Beyond", Asilomar, Pacific Grove, CA. *Cation-induced assembly of nucleosomes: counting the number of ions*

7. 03/2011, Biophysical Society 55th Annual Meeting, Baltimore, MD. *Determination of the Composition of the Ion Atmosphere of Condensed DNA Utilizing Inductively-Couple Plasma Atomic Emission Spectroscopy*

8. 03/2011, Biophysical Society 55th Annual Meeting, Baltimore, MD. *Contrast Variation SANS Study of DNA Packaging in Bacteriophage lambda*

9. 03/2011, Biophysical Society 55th Annual Meeting, Baltimore, MD. *Mapping the Lengths of DNA inside Partially Ejected Bacteriophage lambda*

Distributed Softwares (*available at http://home.gwu.edu/∼xqiu*)

1. X. Qiu, J. W. Thompson, and S. J. L. Billinge, **PDFgetX2***: a GUI driven program to obtain the atomic pair distribution from x-ray powder diffraction data*.
2. X. Qiu, **SASImage***: a graphic interface for quick analysis of SAXS 2D data*.
3. X. Qiu, **xSQfit***: refinement of inter-particle interference function with a graphic user interface*.
4. X. Qiu, **atomSAXS***: population of counterions with electrostatic calculations and calculation of SAXS profiles based on total atomic structures*.

Publications

- Significant Peer-reviewed Journal Articles

  1. X. Qiu, K. Andresen, S. C. Howell, I. Jimenez-Useche, and C. Yuan, *Measuring inter-nucleosome interactions and the roles of histone tails*, **Submitted to Physical Review Letters**, (2012).
  2. X. Qiu, *Heat-induced capsid disassembly and DNA release of bacteriophage λ*, **PLoS One**, in press (2012).
  3. X. Qiu, D. C. Rau, V. A. Parsegian, L. T. Fang, C. M. Knobler, and W. M. Gelbart, *Salt-dependent DNA-DNA spacings in intact bacteriophage lambda reflect relative importance of DNA self-repulsion and bending energies*, **Physical Review Letters**, **106**, 028102 (2011).
  4. X. Qiu, V. A. Parsegian, and D. C. Rau, *Divalent counterion induced condensation of triple-strand DNA*, **Proceedings of the National Academy of Sciences**, **107** (50), 21482–21486 (2010).
  5. X. Qiu, and P. Setlow, *Structural and genetic analysis of x-ray scattering of bacterial spores*, **Journal of Bacteriology**, **191** (24), 7620 (2009).

6. S. A. Pabit, X. Qiu, J. S. Lamb, L. Li, S. P. Meisburger, and L. Pollack, *Counterions more effectively screen dsRNA than dsDNA*, **Nucleic Acids Research**, **37** (12), 3887 (2009).
7. X. Qiu, K. Andresen, J. S. Lamb, L. W. Kwok, and L. Pollack, *Abrupt transition from free, repulsive to condensed, attractive DNA phase, induced by multivalent polyamine cations*, **Physical Review Letters**, **101**, 228101 (2008).
8. X. Qiu, K. Andresen, L. W. Kwok, J. S. Lamb, H.-Y. Park, and L. Pollack, *Inter-DNA attraction mediated by divalent counterions*, **Physical Review Letters**, **99**, 038104 (2007).
9. X. Qiu, L. W. Kwok, H.-Y. Park, J. S. Lamb, K. Andresen, and L. Pollack, *Measuring Inter-DNA Potentials in Solution*, **Physical Review Letters**, **96**, 138101 (2006). (*selected as a science highlight by Synchrotron News at lightsources.org*)
10. H.-Y. Park, X. Qiu, E. Rhoades, J. Korlach, L. W. Kwok, W. R. Zipfel, W. W. Webb, and L. Pollack, *Achieving uniform mixing in a microfluidic device: hydrodynamic focusing prior to mixing*, **Analytical Chemistry**, **78** (13), 4465–4473 (2006).
11. X. Qiu, Th. Proffen, J. F. Mitchell, and S. J. L. Billinge, *Orbital correlations in the pseudo-cubic* O *and rhombohedral* R*-phases of* $LaMnO_3$, **Physical Review Letters**, **94**, 177203 (2005).
12. X. Qiu, J. W. Thompson, and S. J. L. Billinge, *PDFgetX2: A GUI driven program to obtain the pair distribution function from x-ray powder diffraction data*, **Journal of Applied Crystallography**, **37**, 678 (2004).
13. P. J. Chupas, X. Qiu, J. C. Hanson, P. L. Lee, C. P. Grey, and S. J. L. Billinge, *Rapid-acquisition pair distribution function (RA-PDF) analysis*, **Journal of Applied Crystallography**, **36**, 1342–1347 (2003).

- Other Peer-reviewed Journal Articles

14. K. Andresen, X. Qiu, J. S. Lamb, H.-Y. Park, L. W. Kwok, and L. Pollack, *Mono- and Tri-valent Ions around DNA: A Small-Angle Scattering Study of Competition and Interactions*, **Biophysical Journal**, **95**, 287–295 (2008).
15. J. S. Lamb, L. W. Kwok, X. Qiu, K. Andresen, H.-Y. Park, and L. Pollack, *Reconstructing three dimensional shape envelopes from time resolved small angle x-ray scattering data*, **Journal of Applied Crystallography**, **41**, 1046–1052 (2008).
16. G. Campi, Th. Proffen, X. Qiu, E. S. Božin, S. J. L. Billinge, S. Agrestini, N.L. Saini, and A. Bianconi, *Local lattice dynamics in the* $Mg_{0.5}Al_{0.5}B_2$ *superconductor*, **Journal of Superconductivity and Novel Magnetism**, **20**, 505–510 (2007).
17. J. S. Lamb, S. Cornaby, K. Andresen, L. Kwok, H.-Y. Park, X. Qiu, D. M. Smilges, D. H. Bilderback, and L. Pollack, *Focusing capillary optics for use in SAXS*, **Journal of Applied Crystallography**, **40**, 193–195 (2007).
18. E. Božin, X. Qiu, R. J. Worhatch, G. Paglia, M. Schmidt, P. G. Radaelli, J. F. Mitchell, T. Chatterji, Th. Proffen, and S. J. L. Billinge, *Utilizing total scattering to study the Jahn-Teller transition in* $La_{1-x}Ca_xMnO_3$, **Z. Kristallography**, Suppl. **26**, 429–434 (2007).
19. G. Campi, E. Cappelluti, Th. Proffen, X. Qiu, E. S. Božin, S. J. L. Billinge, S. Agrestini, N.L. Saini, and A. Bianconi, *Study of temperature dependent atomic correlations in* MgB_2, **The European Physical Journal B**, **52**, 15–21 (2006).
20. E. S. Božin, X. Qiu, M. Schmidt, G. Paglia, J. F. Mitchell, P. G. Radaelli, Th. Proffen, and S. J. L. Billinge, *Local structural aspects of the orthorhombic to pseudo-cubic phase transformation in* $La_{1-x}Ca_xMnO_3$, **Physica B**, (385–386), 110–112 (2006).

21. X. Qiu, S. J. L. Billinge, C. R. Kmety, and J. F. Mitchell, *Evidence for nano-scale inhomogeneities in bilayer manganites in the Mn^{4+} rich region: $0.54 \leq x \leq 0.80$*, **Journal of Physics and Chemistry of Solids**, **65**, 1423–1429 (2004).
22. P. J. Chupas, S. Chaudhuri, J. C. Hanson, X. Qiu, P. L. Lee, S. D. Shastri, S. J. L. Billinge and C. P. Grey, *Simultaneous local and long range structure determination: application to in-situ studies of the high temperature phase transition of α-AlF_3*, **Journal of the American Chemical Society**, **126**, 4756–4757 (2004).
23. X. Qiu, E. S. Bozin, P. Juhas, Th. Proffen, and S. J. L. Billinge, *Reciprocal-space instrumental effects on the real-space neutron atomic pair distribution function*, **Journal of Applied Crystallography**, **37**, 110–116 (2004). (Times Cited: **19**)
24. S. Derakhshan, A. Assoud, E. Dashjav, X. Qiu, S. J. L. Billinge, and H. Kleinke, *Distorted nets of Ti atoms comprising squares and rhombs in the new binary antimonide Ti_2Sb*, **Journal of the American Chemical Society**, **126**, 8295–8302 (2004).

- Newsletter & Others

25. X. Qiu and S. J. L. Billinge, **PDFgetX2***: a user friendly program to obtain the atomic pair distribution from x-ray powder diffraction data*, Commission on Powder Diffraction Newsletter, International Union of Crystallography, No. 30, 36 (2005).
26. P. J. Chupas, C. P. Grey, J. C. Hanson, J.-Y. Kim, J. Rodriguez, X. Qiu, S. J. L. Billinge, and P. L. Lee, *In-Situ time resolved powder diffraction studies in heterogeneous catalysis; coupling the study of long range and local structural changes*, Commission on Powder Diffraction Newsletter, International Union of Crystallography No. 29, 24 (2003).

Scientific Community Services

- Grant Proposal Reviewing

  Office of Naval Research
  Research Corporation for Science Advancement
  Neutron Scattering Science at the Oak Ridge National Laboratory

- Journal Peer-Reviewing

  Physical Review Letters
  Physical Review E
  Physical Biology
  Journal of Biological Physics
  Proteins: Structure, Function, and Bioinformatics
  Journal of Physical Chemistry
  European Biophysics Journals

- Other Services

1. 12/2011, Abstract Sorter for American Physical Society March Meeting, College Park, MD
2. 05/2011, Career Q&A Panelist, NICHD Fellows 7th Annual Meeting, Warrenton, VA
3. 04/2011, Poster Prize Judge, DC Science & Engineering Fair, Washington, DC
4. 03/2011, Grand Prize Judge, Fairfax County Regional Science and Engineering Fair, Fairfax, VA

5. 12/2010, Abstract Sorter for American Physical Society March Meeting, College Park, MD
6. 03/2009, Session Chair for American Physical Society March Meeting, Pittsburgh, PA

Memberships

Biophysical Society
American Physical Society
American Crystallographic Association
Neutron Scattering Society of America